

Semantic See-through Goggles: Wearing Linguistic Virtual Reality in (Artificial) Intelligence

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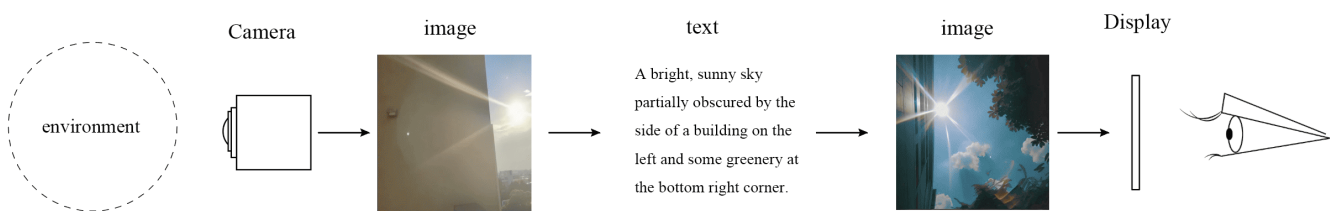


Figure 1: Top: The framework of Semantic See-through Goggles. Bottom: The prototype of Semantic See-through Goggle. One and Multiple Realities.

Abstract

When language is used as a medium for sensory information—as when one describes a scene in words—a kind of virtual reality

emerges: realities projected into the same sentence become virtual/equivalent. We call this Linguistic VR (LVR). LVR is constituted by codes shaped chiefly by majority cultural norms. In today's world—where AI, largely built on linguistic data and processes, is deeply entangled with the everyday mediation of sensory information—it is necessary to critically re-examine the virtuality of this VR.

We propose Semantic See-through Goggles, a system that makes manifest, as a first-person experience, the LVR latent in the linguistic mediation of scenes, enabling intuitive understanding and



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analysis of its properties and issues. The system inserts a serial image-to-text and text-to-image transformation pipeline between the camera and head-mounted display (HMD) of video see-through goggles: the live view is converted into a single line of text, then re-generated as an image, and only this mediated image reaches the wearer’s eyes. We built a prototype and validated its basic properties, followed by a qualitative analysis of wearer experiences. The results suggest that this method enables subjective experience—and thus understanding—of the transformations of environmental information, and of the attendant issues, induced by the linguistic mediation of vision. We also obtained preliminary insights into both AI-driven linguistic mediation of sensory information and problems intrinsic to linguistic mediation itself.

CCS Concepts

• **Human-centered computing** → **HCI theory, concepts and models**; *Virtual reality*; • **Computing methodologies** → *Artificial intelligence*.

Keywords

language, virtual reality, see-through, AI

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1 Introduction

Language functions as a medium for transmitting sensory information. For example, with words, you can show what’s in front of you now to someone somewhere, someday [6, 80]. What appears to the receiver is an image that is not physically “there,” yet possesses enough efficacy to enable description, judgment, and action with respect to it. In this sense, language constitutes a broad form of Virtual Reality (VR): as an apparatus that affords an experience that functionally/virtually corresponds to direct perception of reality.

Viewing language from the perspective of VR foregrounds an intense hypothesis of virtuality: “the set of realities projected into the same sentence are virtual/equivalent.” Analogously, HMD-based VR potentially has hypothesis that the set of realities that projected into the same retina image are virtual/equivalent; more broadly, many forms of VR, projected into the same sense data are virtual/equivalent. At a rational level, people recognize that the scene they imagine from language does not coincide with the scene perceived by the speaker. Yet, in actual interaction, such discrepancies recede into the background under this hypothesis, and communication proceeds. This virtuality can be considered to be supported by codes grounded in the cultural norms of the majority [11, 40]; the selections made during verbalization depend on the value assignments of the linguistic community [8, 53]. Consequently, information that is not put into words not only drops out but is also likely to be rewritten during interpretation into the “safest,” most generic image. The very mechanism that enables language’s high compressibility and creativity simultaneously harbors risks of bias, oppression, and misreading.

Today, AI systems that rely on linguistic data sources and procedures are pervasively and at scale involved in the linguistic mediation of our sensory information [17, 21]. By enabling fast, large-scale mediation, AI not only reproduces the problems of language-based sensory mediation described above, but often amplifies them through uniformity and the loss of non-linguistic cues [10, 48, 66]. Numerous reports have documented AI’s reproduction of biases and forms of oppression inherent in language—particularly regarding gender and ethnicity. Although these phenomena are often discussed as issues peculiar to AI, they should, at a more fundamental level, be understood as extensions of the virtuality inherent to language as a mode of mediation. How, then, are sensory data transformed by linguistic filters deemed “virtual,” what does it mean to perceive the world through images thus deemed virtual, and what does it mean to outsource a portion of that process to AI? These questions call for empirical investigation within an HCI framework.

This study proposes Semantic See-through Goggles, a system that renders Linguistic VR—latent in the linguistic mediation of scenes—into a first-person, action-coupled experience, enabling intuitive understanding and analysis of its properties and problem structure. The proposed system inserts Image-to-Text (I2T) and Text-to-Image (T2I) transformations in series between the camera input and the display of a video see-through HMD. Concretely, the scene before the eyes is first summarized and verbalized into a single line of text (I2T), which then serves to regenerate an image (T2I); the user perceives only this regenerated image while acting in the physical environment. In this way, the chain of verbalization → re-visualization and its information transformations are presented as a primary experience inseparable from cognition and motor action.

The remainder of this paper is organized as follows: Section 2 reviews related work; Section 3 presents the design framework; Section 4 describes the prototype and evaluation procedure; Section 5 reports qualitative findings from a workshop; Section 6 discusses limitations and design implications; and Section 7 concludes with directions for future work.

2 Related works

2.1 Mediation of Visual Information via AI

In recent years, artificial intelligence (AI) has increasingly mediated sensory information, especially in generating and interpreting visual data. Multimodal systems—most notably Vision-Language Models (VLMs) and (in some deployments) LLM-based multimodal models—learn alignments between visual representations and meaning using linguistic supervision and vision-language data, enabling information to be transformed and reconstructed across modalities [42, 65]. This design is clearest in image-to-text [41, 74] and text-to-image models [58, 61]. Even when no explicit conversion is performed, multimodal models often map images into a shared semantic space aligned with language, supporting retrieval and reasoning within that space [34, 56]. The following sections summarize key forms of this linguistic mediation.

2.1.1 Image to Text. Image captioning takes an image as input and generates text describing salient content such as events, objects, and human actions. Earlier pipelines sought to infer high-level

semantics from low-level visual features [7], but the field shifted toward end-to-end learning with encoder–decoder architectures. A landmark model by Vinyals et al. encodes an image with a CNN and generates sentences with an RNN [74], enabling direct learning from large image–caption datasets. Attention mechanisms later improved fluency and accuracy [1, 78]. More recently, VLM pretraining (e.g., BLIP) aligns vision and language in a shared latent space to strengthen generation and grounding [41]. Consequently, image-to-text models have evolved from mapping visual features into words to reconstructing perceptual information within a linguistically structured semantic space [65]. While still below human-level performance, captioning is used in practical settings, including generating captions for images in news workflows [72] and supporting users with sensory impairments [45].

2.1.2 Text to Image. Text-to-image generation takes natural language prompts and produces images depicting described scenes, entities, or actions. Historically, computer image generation often relied on explicit, low-level drawing instructions, exemplified by plotter-based drawing machines [52]. With deep learning, image synthesis shifted toward learning-based generative models. GANs [26] and VAEs [36] enabled realistic synthesis by modeling data distributions, but have been reported to show limited semantic consistency with linguistic inputs [75]. In contrast, recent text-conditioned diffusion models—including DALL·E 2 [57], Imagen [64], and Stable Diffusion [61]—use text embeddings as conditioning signals (often in latent diffusion spaces), guiding generation with linguistic semantics. Accordingly, image synthesis has shifted toward producing perceptually coherent content explicitly conditioned on language representations [42]. Applications continue to expand, such as supporting early-stage ideation in architectural design [55] and assisting animation workflows via automated inbetweening, colorization, and storyboard generation [70].

2.2 Understanding and Addressing Bias in AI

2.2.1 Bias in AI. The rapid advancement of multimodal AI as a medium for visual communication and interaction has intensified calls to critically examine its social acceptance and everyday integration [17, 21]. From both technical and ethical perspectives, a central challenge is bias in model behavior and outcomes. For example, debates around the COMPAS recidivism prediction system highlight concerns that such systems can yield unfair impacts under common evaluation criteria [20]. Likewise, commercial facial analysis systems have shown substantially higher error rates for Black women than for white men [12]. Systematic biases and bias-related failures have also been documented in vision pipelines, including settings involving the depiction of human faces [29]. Together, these findings underline the risk that AI may reproduce—and sometimes amplify—existing social inequities.

A major driver of bias is imbalance in the collection, annotation, and distribution of training data. Because models learn by reproducing statistical regularities in their training sets, social and cultural skews in datasets can be reflected in downstream outputs [5]. Torralba et al. showed that widely used benchmark image datasets (e.g., Caltech101 and PASCAL VOC) contain distinct visual and statistical skews [71]. Bhargava et al. further identified gender and occupational imbalances in COCO Caption that can lead captioning

models to reproduce stereotypes [8]. Comparable patterns appear in language data: Caliskan et al. demonstrated that word embeddings encode socially salient associations (e.g., “male–science,” “female–arts”) directly in the embedding space [13]. On this view, bias is often less an incidental defect than a structural reflection of the data distributions from which systems learn.

Recent work also emphasizes that bias is intertwined with how linguistic semantic spaces are constructed. Because many models encode meaning through co-occurrence statistics, geometric representations can preserve and organize prevailing social value systems. Bolukbasi et al. showed that Word2Vec’s embedding space preserves gender-stereotypical relations such as “man : computer programmer = woman : homemaker” [10]. Marinucci et al. argue that insofar as language models rely on the distributional hypothesis—that meaning is defined by co-occurrence relations [31]—bias is difficult to fully eliminate as a byproduct of linguistic learning [48]. Relatedly, Steed et al. show that social biases can be present in learned visual representations, indicating that such structures can appear in and move across representational domains [66]. More broadly, these issues intersect with the symbol grounding problem [30]: if “meaning” is learned primarily from dataset correlations rather than embodied perceptual experience, model understanding may amount to a reconstruction of a linguistically and culturally mediated world, with its biases carried along [24].

2.2.2 Approaches to Bias in AI. Approaches to mitigating AI bias are commonly categorized by the stage of intervention: (1) modifying training data during preprocessing, (2) intervening in the training process, and (3) addressing bias in outputs post hoc. While these strategies aim to promote fairness, fully automated solutions face fundamental limitations because fairness is inherently context-dependent [5]. Human-in-the-Loop (HITL) frameworks therefore attract attention for reintroducing human judgment and social context into model training, evaluation, and decision-making [77].

At the same time, purely technical “debiasing” can be superficial. Gonen et al. argued that many debiasing methods for word embeddings leave underlying biased structures largely intact (“lipstick on a pig”) [25], and Birhane et al. caution that framing bias as a problem solvable by technical fixes alone can obscure deeper social and historical conditions [9]. Reflecting this, recent work increasingly emphasizes interpretability and methods for making bias visible and diagnosable [19]. Concrete techniques span the pipeline: crowd-sourcing workflows to detect dataset bias [33]; discovering latent bias attributes via generative-model latent spaces [43]; visualizing transformer attention patterns [73]; and output-level inspection by comparing images and captions [35] or prompts and generated images [54]. Despite these advances, much of the literature remains focused on statistical or output-level disparities, leaving open how structural bias in linguistic semantic spaces becomes manifest in human perception and experience.

2.3 See-through Goggles

To address the background and challenges described thus far, we adopt an approach utilizing see-through goggles. Because such devices mediate the relation between human perception and the external world, we hypothesize that they can turn AI bias and

related structural issues into first-person perceptual experiences—reframing them as personally situated problems. This section introduces see-through technology both as (i) the basis of many AR systems and (ii) an experimental apparatus that can transform the act of “seeing through,” and summarizes the foundational knowledge needed for system design.

2.3.1 See-through Goggles as the basis of AR. “See-through goggles” are devices that allow users to see through them, functioning as a medium between perception and the external world. In AR, they are commonly categorized as optical see-through and video see-through [3, 50]. Optical see-through overlays digital information onto the real world via a transparent display; Sutherland’s “The Sword of Damocles” is widely regarded as an early prototype [68], and this approach has informed devices such as Google Glass and Microsoft HoloLens. Video see-through, by contrast, captures the world with a camera, displays it, and composites digital content into the image, offering greater flexibility because the visual scene is fully digitized. Bajura et al. demonstrated such integration by overlaying real-time ultrasound imagery onto a patient’s body while preserving spatial alignment in the user’s view [4]. See-through systems also face depth-related challenges such as accommodation–convergence conflict and latency, yet users can adapt perceptually through visuomotor recalibration [16]. Overall, see-through goggles can be understood as mediating devices that maintain perceptual transparency while enabling controlled modification of visual experience.

2.3.2 See-through Goggles as Experimental Devices. Beyond AR, see-through goggles have been used as experimental apparatuses that reshape “how” one sees through, rather than simply adding or removing information on top of a transparent view. Such devices alter or suppress key elements of vision and observe how perception is reestablished, thereby probing the relative roles of visual components. Stratton empirically demonstrated perceptual adaptation using a prism device that rotated the visual field by 180 degrees [67]. Erismann and Kohler later showed, through long-term studies with inversion and color-filter goggles, that vision–action coupling reorganizes dynamically in response to changed environments [63]. Building on this tradition, Mann’s EyeTap-based notion of Mediated Reality treats mediation as not only adding/subtracting content but also modifying the see-through structure itself [46, 47]. In a related artistic gesture, Schilling’s Viewing Machines explored alternative configurations of see-through (e.g., inversion-like structures and foreground/background reversal), offering a critical imagination of everyday vision [37]. Our Semantic See-through Goggles follow this experimental lineage by allowing only “what remains once it is put into words” to pass through while filtering out other visual information.

2.3.3 Interaction Design with Bystanders. HMDs, including see-through goggles, are typically designed around the wearer’s immersive experience, which often creates an asymmetry between what the wearer perceives and what bystanders can access [27, 60]. Prior work has addressed this by externalizing the wearer’s viewpoint or experience to support interaction: Gugenheimer et al. equipped an HMD with a touch-enabled front display so bystanders could view the wearer’s perspective and intervene in the virtual environment

[28]; Wang et al. projected the wearer’s viewpoint into the physical space using a small HMD-mounted projector, enabling shared view and collaboration [76]. These studies point to design strategies that prevent HMD experiences from remaining private and instead foster shared understanding. Our system extends this line of work by displaying what participants are viewing on external monitors attached to the goggles and/or on surfaces in the exhibition space (e.g., walls).

At the same time, we want to emphasize that the very motivation of this study—to foreground “Linguistic VR”—also carries a gesture that overlaps with these prior projects: an interest in the appearance of the VR wearer’s subjective world, and a desire to open that appearance to others.

3 Design framework: Semantic See-through Goggles

Semantic See-through Goggles is a system that renders, as a first-person visual experience, the Linguistic VR latent in the linguistic mediation of visual scenes. The system serially inserts two transformations—Image-to-Text (I2T; image description) and Text-to-Image (T2I; image generation)—between the video see-through HMD’s head-mounted camera input and its display output. The abstract processing pipeline is as follows:

- (1) *View capturing*: The head camera captures the current view.
- (2) *Image-to-Text (I2T)*: The captured image is described as a sentence.
- (3) *Text-to-Image (T2I)*: An image is generated from that sentence.
- (4) *View presentation*: The generated image is presented on the HMD display.

The wearer is assumed to observe and interact with the environment through this pipeline; accordingly, each processing stage must meet a temporal budget to support continuous perception and action.

Under Linguistic VR’s operational hypothesis that “realities that map to the same sentence are equivalent (i.e., mutually virtual),” the input and output can be treated as equivalent at the level of the sentence that mediates them. However, Semantic See-through Goggles does not merely require such sentence-level equivalence. The output must also be a visually plausible image that could be generated from language alone—namely, an image produced by T2I under the model’s linguistic priors.

4 Implementation

This section details the implementation of the Semantic See-through Goggles, which we address through its system design and model selection.

4.1 System design

The semantic see-through goggles consist of a general head-mounted display (HMD), a camera capable of video capture, and a computer for software processing (Fig.2). Information transfer from the camera to the computer is achieved via a wired connection. In accordance with the proposed framework, the system executes the following processes. First, the system captures real-time camera



Figure 2: The sketch on the implementation of Semantic see-through Goggles. The system utilizes an HMD equipped with a camera. During the image transmission from the camera to the HMD, the two AIs first convert the landscape image into a single line of text and then re-image it.

images and inputs them into a buffer on the computer. Then, the computer processes the buffered images through two AI-powered transformations, generating a new image. The system simultaneously presents the same generated image, at an identical scale, to both the left- and right-eye displays inside the HMD. Through this binocular presentation, the wearer perceives only the adapted image, rather than the original camera image.

While it is possible to implement this experience using a general HMD, adopting an original HMD with inner and outer displays enables the experience to extend beyond the wearer. The outer display of the HMD shows the three types of content: the original image captured by the camera, the single sentence of text converted from that image, and the generated image, without distortion in their combination (Fig.1 lower left). Bystanders not wearing the HMD, who are observing the wearer, can perform various provocative actions while watching the adaptation process displayed on the outer display. This functionality supports smooth communication with bystanders and is expected to result in the experience being collaboratively constructed through their involvement.

4.2 Model Selection

Since real-time capability is essential for this system, we selected the constituent AI models (for image captioning and image generation) based on their capacity for high-speed transformation. For image captioning, we employed BLIP[41], a model pre-trained on a large boosted trap dataset, to caption the scene of a single image in a single sentence. The model is pre-trained on both visual data, offering high-quality visual representation and linguistic data, providing strong language generation and zero-shot transfer ability. For image generation, we use LCMs [44], a model that streamlines the inference process in the Latent Diffusion Model (LDM), to generate images from single sentences. This model enables rapid inference for pre-trained LDMs with a minimal number of steps. We note that both models rely on a linguistic representation space, and we benefit from their widespread academic acceptance.

To achieve both high transformation speed and improved quality, we implemented several additional techniques to refine the conversion process. First, we limited the length of sentences in

image captioning to between 20 and 50 words. In addition, we standardized both input and output images to 640×640 pixels. This limitation on the output format stabilizes the output variance from the model and simultaneously improves processing speed. Finally, for the computationally intensive image generation process, we adjusted the parameters to run the inference process in just four steps. As a result, we successfully reduced the time required for both processes to approximately one second, enabling the system to follow changes in see-through scenery almost in real-time while maintaining image quality.

5 Preliminary Study

This section provides a preliminary survey of the characteristics of transformation within the semantic see-through framework. We conducted this investigation using three approaches, each with a specific intent: first, an evaluation using sentence similarity metrics; second, an evaluation using image similarity metrics; and finally, a qualitative evaluation based on a web-based questionnaire. These evaluations are positioned to verify whether the system can function as a medium for experiencing ‘Linguistic VR’, the semantically reconstructed reality proposed in this paper.

The dataset consistently utilized for this validation is the freely available DET test dataset employed during ILSVRC 2017. ILSVRC is known as a major challenge for large-scale image classification and object detection (2012–2017)[62]. All images were subjected to preprocessing, consisting of center-cropping and resizing to 256×256 pixels to match the target output resolution of the system. We designed the validation to randomly sample images from the source, ensuring the final dataset contains a diverse mixture of subjects, including people, animals, and landscapes.

To evaluate the conversion results of the actively operating system, we aligned the parameters for image captioning and image generation with those used in real-time applications. Accordingly, we maintained the length of the captioned text between 20 and 50 words and limited the image generation inference step to 4. However, we adjusted the output image to 256×256 pixels to maintain consistency with the input image dimensions.

5.1 Linguistic Semantic Consistency

To most directly verify if the system’s transformation function achieves the principle of ‘Semantic Virtual Reality,’ that realities yielding the same words are indeed the same, we investigated its textual consistency using sentence similarity metrics. In simple terms, the system’s function is to convert a captured scene to text and then back into a scene. However, the current implementation is not a symmetric process, as it relies on distinct models for image captioning and image generation. Although some information is inevitably lost during this transformation, the essential meaning should be retained if the system operates as intended. This evaluation therefore serves as a validation of the fundamental requirement that the system must be able to reconstruct the semantic structure of reality through linguistic transformation.

Method. We used multiple sentence similarity metrics to investigate whether semantic consistency is maintained after linguistic transformation. The procedure involved captioning the pre- and

post-transformation images with the same model and then quantitatively comparing the similarity of the two resulting sets of sentences. As a baseline for comparison, we also calculated scores using randomly combined sentence pairs. This evaluation enabled us to quantitatively examine what was linguistically preserved and lost, thereby validating whether the system can reconstruct the semantic structure of reality through linguistic transformation.

We used four metrics for evaluation: (i) TF-IDF, (ii) Word Mover's Distance (WMD), (iii) Universal Sentence Encoder (USE), and (iv) Sentence-BERT (SBERT). TF-IDF scores the importance of words based on their frequency of occurrence and inverse document frequency, assigning higher scores to words that are less frequent across the documents. WMD evaluates semantic similarity between sentences by calculating distances in a word embedding space[38]. USE utilizes a transformer-based architecture to embed entire sentences, capturing their meanings more effectively than word-level embeddings[14]. SBERT builds on BERT's architecture to encode sentences individually, capturing their nuanced meanings[59]. For word-based comparison metrics like TF-IDF and WMD, preprocessing involves stop-word removal and tokenization. Cosine similarity is used to measure similarities in TF-IDF, USE, and SBERT by assessing the angles between embedded vectors. For WMD, similarity was calculated by taking the reciprocal of the distance; specifically, the similarity is 1 when the distance is 0, and it approaches 0 as the distance increases. For comparison, 2,500 pairs of sentences captioned from images before and after conversion were used. Phrases and words that are continuously repeated in the sentences were removed during a preprocessing step, leaving only one since they are noise in the comparison.

Table 1: Statistics for the linguistic similarities

Indices	Ave _p	Std _p	Ave _R	Std _R	t value	p value	Cohen's d
TF-IDF	0.38	0.18	0.19	0.10	51.75	<0.001	1.01
WMD	0.60	0.08	0.51	0.04	55.44	<0.001	1.09
USE	0.58	0.19	0.26	0.13	81.21	<0.001	1.59
SBERT	0.57	0.20	0.13	0.13	98.65	<0.001	1.93

Result. The distribution of the similarity of the four similarity indices, TF-IDF, WMD, USE, and SBERT, in the Paired Condition (Correspondence Condition) and Random Condition (Random Condition) is shown in Figure 43. Correspondence t-tests were conducted for each indicator, and the results are shown in Table 11. Scores for all indicators tended to be higher in the Paired Condition than in the Random Condition, and statistically significant differences were found ($p < 0.001$). The effect size (Cohen's d[15]) was also large, with Cohen's d greater than 0.8. These results suggest that the image transformation algorithm retains a certain degree of the original image features that each indicator captures in the transformed image.

A comparison of absolute similarity values across the measures showed that TF-IDF had a low average similarity of 0.38 in the Paired Condition, whereas WMD, USE, and SBERT exhibited moderate similarities of 0.60, 0.58, and 0.57, respectively. The higher similarity observed for WMD compared to TF-IDF suggests that, while the algorithm does not maintain lexical-level completeness as captured by TF-IDF, it is capable of preserving the semantic

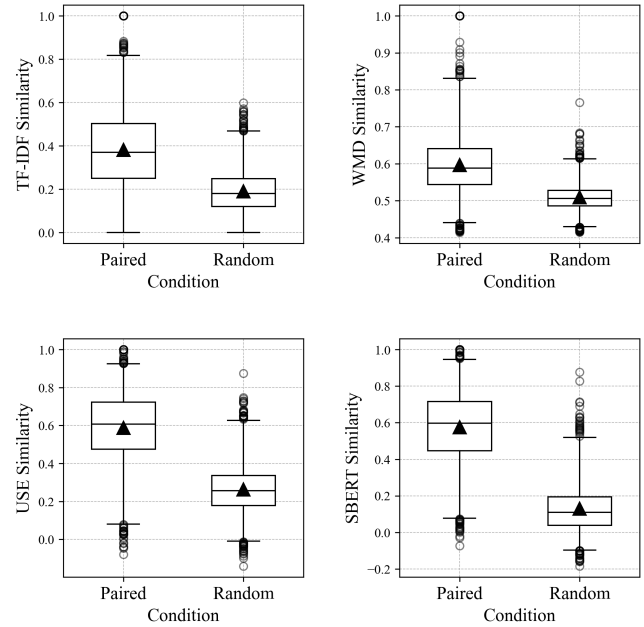


Figure 3: Comparison of linguistic similarities using different similarity indices.The top left figure shows the distribution for each condition in TF-IDF, the top right shows WMD, the bottom left shows USE, and the bottom right shows SBERT. The vertical axis is not aligned in order to see the distribution. Outliers are defined as values in the 99th percentile.

similarity between words, as demonstrated by WMD. Furthermore, the moderate similarity scores from USE and SBERT indicate that the transformation algorithm retains a certain amount of higher-order semantic information embedded in the sentences. Notably, the effect size increased progressively from TF-IDF to SBERT, with values of 1.01, 1.09, 1.59, and 1.93, as the indices became more semantic. This trend suggests that the transformation algorithm is more effective at preserving higher-order semantic information than lexical-level meaning.

5.2 Visual Semantic Consistency

We investigated the general properties of the system's transformation by considering its function as an image-to-image filter. In simple terms, the system captures an image, converts it into text, and then generates a new image from that text. As a result, the input and output images are expected to share the linguistic meaning of a single sentence and visually represent their common features. Although an AI-mediated transformation occurs internally, the process can be treated as a simple image filter when considering only the input and output. We therefore analyzed the characteristics of the system's function by comparing the images before and after this filter was applied, using multiple, distinct types of image similarity metrics.

Method. We analyzed the structural tendencies of visual information during the transformation process by comparing input and

output images using multiple, distinct similarity metrics. First, we generated input-output image pairs by converting dataset images to captions and back again, after which the resulting pairs were scored. Next, for comparison, we compared the scores from this procedure to those calculated for random image combinations. While the validation in Section 5.1 tested for consistency, the present validation is an analysis of the transformation's characteristics. Through this analysis, we can characterize from an engineering perspective how our system re-projects the perceptual information of reality.

We used four metrics: (i) Histogram Intersection (HI), (ii) Scale-Invariant Feature Transform (SIFT), (iii) AlexNet-based Learning Perceptual Image Patch Similarity (LPIPS_A), and (iv) Transformer-based Learning Perceptual Image Patch Similarity (LPIPS_T). HI is a method used to assess the visual similarity between color histograms of different images by calculating the overlapping areas of their histograms. In this case, the calculation is performed in the CIELAB color space, a color model designed to reflect human visual perception[39]. SIFT detects local image features that are invariant to scale and rotation, generating a feature vector. LPIPS (Learned Perceptual Image Patch Similarity) is a method that employs pre-trained deep neural networks to extract features and is believed to offer an evaluation of images that closely aligns with human subjective perception[79]. In this study, we utilized both AlexNet and Transformer models for LPIPS. It is important to note that, unlike other indices, a lower LPIPS score indicates higher similarity between images. For each comparison, 2,500 pairs of input and output images were used.

Table 2: Statistics for the visual similarities

Indices	Ave _P	Std _P	Ave _R	Std _R	t value	p value	Cohen's d
HI Similarity	0.45	0.13	0.39	0.13	19.40	<0.001	0.39
SIFT Similarity	0.45	0.01	0.45	0.01	9.65	<0.001	0.19
LPIPS AlexNet	0.64	0.06	0.69	0.06	-39.43	<0.001	-0.79
LPIPS Transformer	0.67	0.05	0.71	0.06	-37.56	<0.001	-0.75

Result. The distribution of the similarity of four similarity indices, HI Similarity, SIFT Similarity, LPIPS_A, and LPIPS_T, in the Paired Condition (Correspondence Condition) and Random Condition (Random Condition) is shown in Figure 54. We confirmed that the relationship between the Paired and Random distributions varies depending on the metric utilized.

The corresponding t-test results for each index are provided in Table 22, where statistically significant differences were observed for all indices ($p < 0.001$). For HI Similarity, the analysis yielded a Cohen's d of 0.39, with a mean similarity of 0.45 in the Paired Condition, indicating a moderate effect size. This suggests that the transformation preserves color distribution features to a certain extent. In contrast, SIFT Similarity exhibited a similar mean similarity, but the difference from the Random Condition was minimal, with a low Cohen's d of 0.19. This indicates that many local features, such as edges and corners, were lost during the transformation. In contrast, the analysis of LPIPS_A and LPIPS_T showed relatively high average scores in the Paired Condition, at 0.64 and 0.67, respectively. Cohen's d values were also large, at -0.79 and -0.75, respectively. Since higher scores in the LPIPS index indicate lower similarity, these absolute scores confirm that the conversion process does not

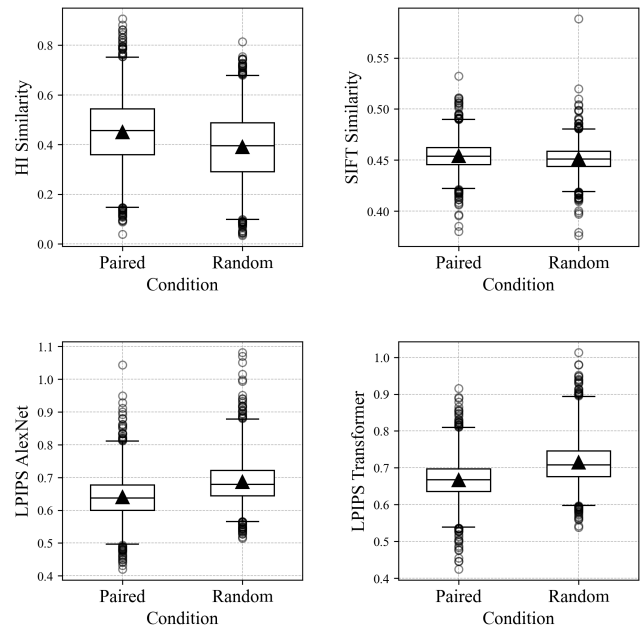


Figure 4: Comparison of visual similarities using different similarity indices. The top left figure shows the distribution for each condition in HI similarity, the top right shows SIFT similarity, the bottom left shows LPIPS_A, and the bottom right shows LPIPS_T. The vertical axis is not aligned in order to see the distribution. Outliers are defined as values in the 99th percentile.

effectively preserve visual and higher-order semantic information as measured by LPIPS. Nonetheless, the magnitude of the effect sizes suggests that the transformation algorithm has a statistically significant impact on preserving some aspects of this information. Based on these results, the image transformation in Semantic See-through Goggles alters local features, while color information tends to be preserved. Additionally, the transformation was also found to have a consistent effect on visual and higher-order semantic information.

5.3 What is being preserved/lost?

This section describes a qualitative analysis of what is retained and lost during the transformation process, based on subjective human evaluation. In this system, an input image is converted to an output image via text. As a result, fundamentally only the textual content is preserved, while other information is either lost or replaced. While Section 5.2 investigated these characteristics with quantitative metrics, the present validation utilizes human perception as its standard.

Method. Subjects were recruited using an online platform, and a web-based questionnaire was administered. Gender, age, and nationality were collected as basic demographic information for the subjects. Participants were asked to observe 200 pairs of input and output images for 10 minutes and then answer the following two

questions. Note that all participants were unaware of our algorithm, i.e., they answered the questions by looking only at the input and output of the images. A total of 24 respondents participated in the study, with ages ranging from 22 to 25. Of the respondents, 20 were male and 4 were female. Regarding nationality, 23 were Japanese, and 1 was Chinese.

Table 3: Item Statements

Item	Statement
Q1	What do you consider to be preserved in the image before and after conversion?
Q2	What do you consider to be lost in the image before and after conversion?

Result. The following elements were extracted from the answers. Participants’ natural-language responses provide qualitative insight into which elements of visual information are preserved or lost through AI-mediated linguistic mediation. Because participants’ intentions in selecting particular terms were not always explicit, we intentionally avoided premature abstraction and instead presented the observed vocabulary in a comprehensive, unaggregated form.

Q1. Situation / Category / Overall theme / Image theme / Subject category / Approximate genre / Main message / Concept / Abstract conceptual information / Main motif / Center-focused subject / Main image / Attention target / Subject of focus / Subject’s objective attributes / Subject characteristics / Feature of the subject / Main subject / Color scheme / Overall color tone / Color tones / Colors / Appearance of colors / Elements in the image / Objects included / Present elements / Main elements / Material / Object information / Object’s category / Biological entities / Presence of living beings / Type of beings / Action / Composition / Layout / Pose of people or animals / Situation / Setting / Scene / General content / Number of objects / Approximate number of objects / Distance from the subject / Spatial relationship with the object / Gender / Gender of entities depicted / Location / Place / Image / Impressions conveyed by the image.

Q2. Details / Margins / Realism / Photo-likeness / Accuracy / Fine details / Clarity / Background / Text / Numerical information / Identity / Overall saturation / Detailed characteristics of the subject / Main subject and background / Specificity / Actions / Movement / Dynamism / Color tone / Saturation / Composition / Image focus / Facial expressions / Race / Specific features of the subject / General theme / Low-importance conceptual information / Concrete information with specificity / Realism / Sense of reality / Style / Time concept / Sequence / Subject’s personality / Photographer’s intention / Position / Posture / Meta-information about the photo / Overall tone / Human expressions / Presence of reflections or additional elements in the image.

Even from the vocabulary distribution in Q1/Q2 alone, the transformation tends to preserve the structural and contextual semantics of the scene (situation/category, main elements, overall color tone, and approximate counts) while losing details that convey individual details. For example, in the case of human-related information, broad attributes such as gender tend to be preserved, whereas

identity, facial expressions, and personality, which are the more individuating aspects, are more likely to be lost. Similarly, for spatial information, while relative spatial relations and overall color tone are retained, metric geometry (position/style) and photographic fidelity (saturation/realism) are often lost, suggesting a resolution gap between coarse-level preservation and the loss of fine-grained information. In addition, while action and pose of people or animals are preserved, movement, sequence, and the time concept are lost, indicating a break in the continuity of motion and revealing that this implementation fails to retain temporal information.

6 User Study

This section reports an exploratory user study combining in-situ behavioral observation and post-experience interviews to examine how people perceive and act when viewing physical reality through the proposed *Semantic See-through Goggles* (Fig. 5). We position this study as a qualitative investigation aimed at characterizing first-person experiences and interaction patterns under real-time I2T→T2I mediation, rather than estimating population-level effects.

Given the bespoke hardware setup, safety supervision, and the time-intensive in-situ protocol, we prioritized depth of observation and rich interviews over breadth. Accordingly, we recruited nine participants (6 men, 3 women; aged 23–35, $M = 25.0$, $SD = 3.77$). None had participated in the preliminary experiment. We iteratively analyzed the data and observed diminishing returns in substantively new experiential themes.

All participants were informed about the study and the system before the demonstration and provided consent to take part. No participant self-reported physical disabilities relevant to the task.

Setting and apparatus. Participants interacted in a shared indoor space within a typical university laboratory. We used a 6 m × 6 m area inside an 8 m × 15 m room; the ceiling lights were on throughout. The space contained various everyday objects (e.g., desks, chairs, fruit, cooked food, tableware, ornamental plants, balloons). Two researchers were co-present as experimenters/confederates for supervision and facilitation, and 1–3 other participants were occasionally co-present while waiting for their turns. The session was supervised for safety; we removed hazardous items, monitored participants’ walking and reaching, and intervened as needed to prevent falls or collisions. Eating was permitted, but in practice only the participant who had prepared the food ate during the session.

Procedure. Each participant used the system for at least 10 minutes (up to 30 minutes as desired). They were encouraged to move freely, visually explore, and perform simple interactions with nearby objects (e.g., locate and grasp an item). After use, the first author conducted a brief face-to-face interview (duration: ~15 minutes) following a semi-structured guide (topics: overall experience; sense of “see-through”; interaction; perceived bias; comparison with ordinary vision and with language-based communication). No participant declined participation or interview.

Data collection. Two forms of data were collected: (1) field notes and video of overt behaviors (camera vantage points: a single hand-held camera operated by the experimenter, following each participant to capture their actions; consent scope: recording permitted for internal research analysis only, with anonymized still frames (faces



Figure 5: The workshop view. Participants wear Semantic See-through Goggles and observe, walk, and interact with the environment involving various objects and people. Participants who are not wearing the goggles can see what the wearer is looking at and what text is being mediated via a front-facing display embedded in the goggles.

blurred) allowed for academic publication; no public release of raw video/audio), and (2) post-experience interview comments labeled by participant ID (P1–P9). Interviews were audio-recorded and transcribed using *faster-whisper* with subsequent manual corrections. We did not collect personally identifying information beyond the demographics reported above; transcripts and quoted comments were anonymized. Quotes are translated from the original interview language into English; edits were limited to grammar and readability while preserving meaning.

Analysis. We conducted an inductive, iterative thematic analysis of interview transcripts and observation notes. Primary coding was performed by the first author, followed by review and refinement with the second author through discussion-driven consolidation. Codes were developed bottom-up in response to the data and subsequently grouped into higher-order themes.

6.1 Results and Discussion

We first describe four recurring behavioral episodes that repeatedly surfaced during observation, then present thematic analyses from the interviews. Together they characterize how participants (i) treated the mediated view as “see-through” enough to support basic action, (ii) formed a first-person, affect-laden grasp of AI adaptation/bias, and (iii) extended reflection beyond AI to language-mediated perception more generally. Throughout, we treat the

prototype as a proof-of-concept to empirically foreground what we call *Linguistic Virtual Reality*.

6.1.1 Symbolic behaviors.

Attending to people. The system often rendered people in concise textual terms (e.g., “a man with black hair standing”), which were then visualized into a stereotyped appearance. Most participants were surprised that gendered and racialized features tended to default toward specific archetypes and that identities could fluctuate across frames. For example, a slender, long-haired Asian man was alternately visualized as a white man or a white woman in the wearer’s view. Despite these shifts, participants could still infer counts and coarse attributes (e.g., glasses on/off, presence/absence, the approach of a new person).

Locating, grasping, and eating objects. Although much spatial information was lost once the view was converted into sentences, most participants were able to interact with the environment using only the information that remained in the mediated text. For example, in one case an apple appeared in the field of view when a participant turned around and looked down. Not knowing the exact position, the participant moved their face closer while extending their hand. When the apple disappeared, the participant slightly turned their face back in the opposite direction of the movement at that moment, and the apple reappeared. By iterating this feedback, the participant was able to take it in hand and then ate it. This

suggests that, for the participant, the scene appeared as something the body could act upon through the system (and even ingest as part of oneself).

When performing an action such as picking something up, the arm appeared in the image; however, that arm did not necessarily emerge from the foreground (i.e., from the participant's own view-point). In the prototype's image captioning, sentences are typically narrated in the third person. This reflects the tendency of many image-captioning models to describe overall content in order to balance caption readability, informativeness, and the risk of information overload [51]. For example, when expressing in text that the apple in front is being touched, an image-captioning model would likely verbalize it as "A person is touching an apple." However, for the user, "I am touching an apple" would be more appropriate. Our field of view is first-person, yet the textual description is rendered in the third person; as a result, images with an unstable point of view can be produced successively. This instability was also amplified by the prototype's frame-to-frame variability (see Sec. 4). Nevertheless, one could envision a Semantic See-through Goggles system that maintains consistency of person reference by incorporating external intervention or an auxiliary model to inject person information—for example, converting "A person" into "I."

Looking up. Most participants preferred the action of looking up. Because the view was converted into sentences, the images in participants' fields of view did not contain direct information about the original positions of objects in the environment or the angle of gaze. However, when they looked upward, expressions such as "wall and ceiling" and "ceiling" appeared in the mediated text, and the image in the field of view gradually shifted to an upward perspective, as if the system were sensing head orientation. This can be regarded as an instance in which bodily information implicitly embedded in linguistic expressions becomes perceptually available through the mediation pipeline.

Behavior of co-present others. Because a portion of the wearer's mediated view was visible to others in the space, co-present people could see it and were prompted to act. Most participants observed or engaged in behaviors that deliberately steered oneself or the scene toward stereotypes so that such meanings were preserved in the output. For example, when a wearer looked at someone's face, that person actively made expressions clear (e.g., smiling or frowning). We also observed attempts to induce interpretations that differed from the actual situation—for instance, adding a mustache with black tape to make oneself appear older, or performing stereotypical gestures of a desired gender.

6.1.2 Thematic Analysis of Interviews. We organize interview findings into three themes: (1) *See-through as an actionable first-person view*; (2) *Subjective understanding of AI adaptation/bias*; and (3) *Reflection on linguistic mediation beyond AI*. Quotes are labeled P1–P9. Across themes, participants' accounts repeatedly negotiated a tension between *actionability* (the view can be used to do things) and *indeterminacy* (the view is unstable, abstracted, or stereotyped).

Theme 1: See-through as an actionable first-person view. Most participants described the mediated view as "see-through enough" to support basic orientation and action, even though it did not preserve fine-grained spatial detail. Their sense of "see-through"

did not depend on visual fidelity but on whether the mediated view maintained a usable coupling with their own exploratory movements (e.g., looking up; turning; reaching), as also reflected in the recurring behavioral episodes above.

Participants reported that the system conveyed coarse environmental structure (people, shelves, ceiling) and that head/gaze movements often produced predictable shifts in what appeared.

P4: It gives a rough idea of the environment, such as a person in view or a bookshelf. It is interesting that even without directly sensing direction, when I look up, I can see the ceiling.

Several participants characterized the experience as distinctly first-person—not because the imagery was stable, but because the perspective remained tied to their bodily position and movements.

P8: When wearing the goggles, the experience became what could be called first-person.

P9: Even though the image changes rapidly, the sense that I am looking from this point of view—that it is a first-person perspective—is maintained.

This first-person actionability was described as qualitatively different from merely observing another wearer's external display: wearing the system turned the mediated view into an actionable constraint on attention and behavior.

P4: Watching someone else's headset display was a different experience from wearing the headset myself. When I wore the goggles, it felt like a copy of information from the real world, and I found myself focusing more on the content and getting information from the environment.

At the same time, participants emphasized the cost of actionability: because directional and metric information was compressed into coarse linguistic cues (e.g., "left," "right," "next to"), acting required additional inference, bodily probing, and reliance on haptics.

P7: The ball was right in front of me in terms of gaze, but I felt a strange frustration because it was not actually reflected there as an image. I somehow sensed it must be in that direction, and relied on the sensation of my hand for the rest.

P9: When I picked up the ball, I sort of sensed the direction and reached for it. The moment I found it, the distance between the camera position and my ball was all connected.

Several participants further reported a pragmatic adaptation: when they focused on acting, they became less sensitive to instability in texture/background and instead attended to what the system consistently preserved as "action-relevant."

P1: Sometimes, when I try to move based on the goggles' view, the fine colors and shapes stop bothering me.

P7: The goggles highlight what you would normally attend to if you were there. Once you get used to it and narrow down what you pay attention to, it becomes easier to move around.

Yet this adaptation was accompanied by a recurrent epistemic disturbance: when the view remained usable but no longer felt like

ordinary perception, participants asked what, exactly, the mediated images are—a concrete thing, a concept, or something in between.

P6: I see a book here. I see the concept of a book, but it is not tied to a specific book. It's an image, so it's not a concept?

P1: What are we looking at here? It clearly refers to reality, but it is not reality. It feels neither concrete nor abstract...

In short, “see-through” was experienced not as faithful visual reproduction but as a *practical coupling* between bodily movement and the mediated view. At the same time, what this view mediated appeared layered between the concrete and the abstract, the real and the unreal—prompting recurring uncertainty about what, exactly, was being seen.

Theme 2: Subjective understanding of AI adaptation/bias. Most participants initially framed their discomfort as an “AI problem,” but their descriptions extended beyond demographic stereotyping. They highlighted (i) demographic and aesthetic defaults (beautification, stereotyped bodies), (ii) salience/segmentation decisions (what is retained vs omitted; what counts as one thing), and (iii) reconstructive interpretation when text is turned back into images. Experiencing these transformations from a first-person viewpoint often elicited affective responses such as fear, unease, and disgust.

P3: Is this what the AI sees?

P1: I felt that I was implicitly sensing algorithmic bias. Seeing it as my view made the bias I knew as knowledge manifest itself in a different way.

A recurring thread was that “bias” felt less like a static distortion and more like an active process that decides what becomes meaningful (and thus visible) in reconstruction.

P4: I was afraid that if there were one or two people, they would be individually recognized and appear as such. But if there were more, they would be grouped together as “several people.” The fear is not only about the transformation of the objects themselves, but also about the process by which “what becomes one thing” is determined.

Several participants emphasized “beautification” as a concrete, first-person manifestation of this adaptive process.

P5: Various things we see tend to be beautified. For example, my own face and body looked beautified.

Others articulated a stronger objection to the system’s autonomy in summarizing and interpreting their view, especially at the text-to-image step.

P6: I am afraid. I don't like reality being rewritten. I dislike how it takes the liberty to decide what I am seeing, extract key points without my permission, and interpret them on its own. I also felt how strongly interpretation intervenes when images are generated from text.

Several participants associated the prototype’s temporal instability (see Sec. 4) with “dreamlike” perception: concepts persisted while details jumped.

P6: It felt dreamlike. The concept remained, but the details lacked consistency and the jumps were close to the experience of a dream.

Not all participants sought immersion. Some treated the system as an object of playful inference, enjoying the act of reasoning about the intermediate sentence that might have generated the current image.

P7: I was looking at the view, trying to figure out what kinds of words were being used.

P6: I enjoyed the feeling of indirect reasoning—thinking from what actually exists to the sentence, and then to the landscape produced from that sentence. The two views are different but share categories, and it was interesting to deduce the differences.

In short, participants experienced AI “bias” not merely as skewed attributes but as adaptive rules that automatically extract what matters, decide what counts as “one” object/event, and reconstruct the view through interpretation (including averaging and beautification) in a first-person pipeline—eliciting responses that ranged from fear and disgust to playful inference.

Theme 3: Reflection on linguistic mediation beyond AI. As the interviews progressed, many participants shifted from treating the experience as specific to AI toward recognizing a more general structure: mediating a view through linguistic expression. In this reframing, the system’s reductions and reconstructions were understood as an intensified instance of what also happens in everyday verbal communication—selecting, omitting, and relying on shared background knowledge.

P7: At first, I thought this was a subjective experience of the AI process. But then I began to realize it is no different from what happens when we communicate verbally. When I share my experiences, I inevitably condense them into short sentences, and in doing so I'm also cutting down information. In that sense, I feel what I am doing is essentially the same.

Some participants suggested that while the final “texture” differed, the textualization step itself felt close to ordinary human description.

P5: I think the result would be more or less the same whether it was done by a person or by an AI. The final texturing is different, though. Especially the textualization might be almost the same.

Participants also reinterpreted stereotyped outputs as partly a linguistic phenomenon: in everyday descriptions, many demographic attributes are left implicit rather than explicitly named, and such omissions shape how a listener reconstructs a scene.

P9: Everyone turned into Westerners. I think it is due to the assumptions of this model. This appears to be an AI problem, but it is a language problem. When describing a scene domestically, you wouldn't explicitly use the words “a Western man.” In verbal communication, it matters not only what becomes words, but also what shared background knowledge is omitted.

Several participants related this to familiar experiences where linguistic instructions are converted into embodied action (e.g., “right/left” guidance), highlighting that linguistic mediation already entails an internal reconstruction process.

P9: It is similar to the feeling you get when you are doing a suika-wari game (watermelon splitting). People around you say “right” or “left,” and that textual information is converted in your head and...

Some participants arrived at this insight not during their own use, but when observing the external display of others and noticing the mediated text.

P2: At first I attributed the unpleasantness entirely to characteristics of the AI. But when I saw the words mediating other people’s experience, I realized that it was also a characteristic of verbalization itself.

Others extended the same logic to memory: putting experience into words can compress and transform what is retained.

P5: Visual memories from childhood can feel as if they disappear once they are put into words. That is how we make memories smaller.

Finally, several participants expressed that this reframing increased the perceived complexity of the problem: if the issue is structural to linguistic mediation, it cannot be resolved by “fixing the model” alone.

P2: It seemed far more daunting to see it as a general language problem than to see it as an AI problem. I had thought it could be solved to some extent by eliminating model biases and adapting it to human society.

In short, participants reframed the experience beyond AI toward linguistic mediation as a structural process that compresses a view into language and reconstructs it through omission, shared priors, and embodied conversion—making the phenomenon feel more general, and harder to “solve” by model tuning alone.

7 Discussion

This section discusses the device of Semantic See-through Goggles and the conceptual possibilities it suggests. First, we articulate Semantic See-through Goggles as a methodological device for examining AI’s linguistic mediation in a first-person setting. Next, we introduce the concept of “Linguistic Virtual Reality” suggested by this framework. Finally, we discuss limitations of both the prototype and the broader Semantic See-through framing.

7.1 SSTG as a method for understanding AI-mediated linguistic mediation

The proposed system compresses information into a single line of text once the view is in the process of reaching the display image from the camera image. This allows participants to subjectively experience what is lost in the representation of the world as linguistically mediated by AI, and what biases may be introduced to compensate for what is lost. In fact, the workshop and its analysis allowed participants to experience—sometimes with surprise and a sense of conviction—various characteristics associated with AI systems and their failure modes. Here, we reiterate the unique advantages that this framework has as a tool for reflecting on the

“perception of AI” (i.e., what the system appears to notice, retain, and reconstruct), in parallel with its role as a tool for interrogating the AI-mediated representation pipeline.

7.1.1 Subjectivizing the problem. At first glance, many of the AI characteristics surfaced through the Semantic See-through Goggles experience resemble points that have already been discussed in prior work—for example, participants’ observations that abstraction and averaging can reflect dataset/model biases. However, experiencing these issues while observing the world in which one’s body is situated and acting on the basis of that mediated view produced a strange sensation and a form of bodily and affective understanding. This suggests that the tool can help those who lack knowledge about the seriousness of AI-related issues to grasp them quickly and vividly. It also suggests an opportunity for AI researchers to rethink the problems they tackle in everyday work by mobilizing their own perceptual and bodily senses.

At the same time, it is important to treat the scope of what is “subjectivized” with care. As noted in the introduction, our current prototype focuses on an explicit language-mediated pipeline (image captioning and text-conditioned image generation). Other forms of AI-mediated perception may involve semantic information only implicitly or at different stages. Accordingly, it is best understood as a method for experiencing one concrete class of language-mediated AI perception rather than as a general experience of “seeing the world as AI.”

7.1.2 Generalizing problems as a matter of semantic/linguistic mediation. As participants themselves came to realize during the experience, the issues understood and felt through this tool also relate to the general problem of scenes being mediated through language. For example, suppose you tell a friend, “I met a couple at a café yesterday.” The listener may implicitly imagine a “default” couple (e.g., standard build, majority race/sexuality for that context) unless further specified. In this sense, verbal mediation of a scene already contains a structure similar to AI-based image captioning and image generation: when visual information becomes text, selection and segmentation are shaped by the biases of a subject and/or a community; when a recipient reconstructs a scene from that text, reconstruction can again be guided by those biases, often yielding an averaged or beautified image.

The proposed method makes this misalignment available as one’s own view in real-time and in first-person. As our workshop suggests, this can be both surprising and disconcerting. Moreover, as participants also noted, such mediation occurs not only between people but also in memory recall, where a similar phenomenon can be described as linguistic mediation of a view from oneself to oneself. By allowing participants to simultaneously experience (i) a problem that linguistic animals cannot fully escape and (ii) a problem of AI-driven linguistic mediation in the form of a see-through HMD, the method invites recognition of AI-related issues not merely as transient technical defects but as problems tied to how intelligence organizes the world under meaning.

7.2 Linguistic Virtual Reality

When language, as a medium that conveys the sensory information of reality, provides an experience that is effectively (i.e., functionally) correspondent to directly perceiving reality, it can be understood as a system of virtual reality in the broad sense. We call VR in language “Linguistic VR.”

Virtual reality is always accompanied by the question “What is the virtuality of reality?”—namely, in what sense is one experience virtual (i.e., equivalent) to another experience of reality? For many VR devices, the general answer is “identity of the stimuli delivered to the sense organs.” For vision, for example, virtuality is the identity of the image projected onto the retina; for audition, it is the identity of the vibrations of the eardrum. In other words, there is a working assumption that “realities that yield the same sensory stimulation are mutually virtual” [49]. In Sutherland’s view, a mechanism that could arbitrarily control the molecules of surrounding objects was the ultimate display precisely because it could reproduce, for every sense organ and in any situation, the same stimuli as reality [69]. From wall-type systems like the CAVE to displays placed in front of the eyeballs, and more recently to devices that project lasers directly onto the retina, these are, in essence, all attempts to reproduce in the sense organs the stimuli that would have been sensed when directly perceiving the target reality.

Linguistic VR answers this question by grounding virtuality in the identity of the sentence obtained when the experience is put into words. In other words, here we adopt the working assumption that “realities that become the same sentence are mutually virtual.” When we see or hear something, we can put that perceptual experience into language; we can then transmit that language to others or to ourselves and imaginatively re-enact the perceptual experience. In this process, information is dropped; that is, there exist multiple sensory situations that yield the same sentence. Just as, when a three-dimensional object is mapped onto the retina, depth information is lost and multiple distinct 3D shapes can produce the same retinal image, so too can multiple perceptual states be compressed into the same verbal description.

Figure 6 sketches a comparison between sensory virtual reality in vision (e.g., HMD-like VR and projection/perspective-based equivalence in retinal imaging) and linguistic virtual reality. (1)–(3) are sensory equivalents of each other, since the image on the retina does not change. However, (4) is not equivalent because the image on the retina is larger, and (5) and (6) are not equivalent because not only the scale but also the visual image changes significantly. In linguistic virtual reality, however, we consider the equivalence when the visual image is further transformed into language. When the visual image becomes language, it can be thought of as when it becomes a sentence in the mind, or when it is conveyed to another person as a sentence, etc. Here, (2)–(5) are equivalent to each other because they are “trees” in terms of words, although they are different in size and type. On the other hand, (6) is not equivalent because it is a “tower.” Furthermore, it is important to note that (1) and (2) form the same retinal image, but (1) is not linguistically equivalent because it is mistaken for a person due to some influence, and thus becomes a “person.” Thus, the relationship between sensory virtual reality and linguistic virtual reality is not a simple one in which the latter is less virtual.

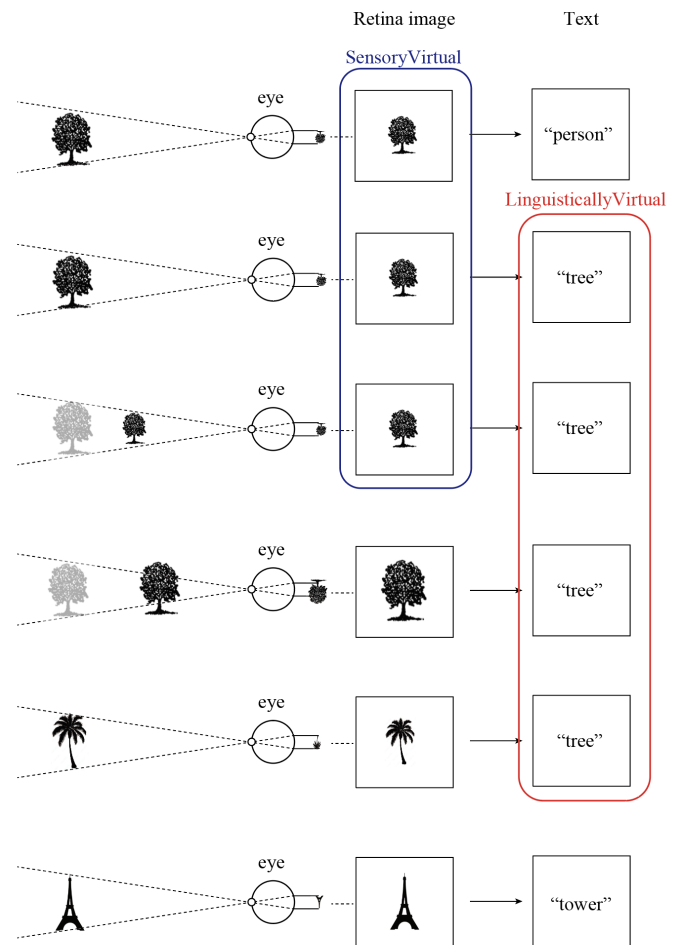


Figure 6: Linguistic/Semantic Virtual Reality.

Our project demonstrates how this kind of linguistic equivalence can be superimposed on Sutherland’s HMD, one symbol of VR in the usual sense, so that it can be seen as a virtual reality. We believe that this idea itself, as a conceptual device, has the potential to connect the sociological study of linguistic communication with the study of ethics in AI and the study of virtual reality.

7.3 Limitations

Here we describe the prototype limitations and envisioned variations to discuss the implementation possibilities of the Semantic See-through framework. A central limitation of the current prototype lies in its dependence on the specific I2T and T2I models employed, which were chosen to favor real-time performance over maximal fidelity. As a result, some observations may change with future advances in AI models. Nonetheless, the prototype is sufficient to instantiate Linguistic Virtual Reality as an embodied, first-person proof-of-concept experience.

7.3.1 Prototype limitations. The most significant limitation is the need to improve real-time performance. The system’s core process involves textualizing a captured camera image with an image captioning model, and then generating a new image from that text using an image generation model. This is computationally intensive, and the system’s defining feature—transforming the visual field in near real-time—is currently realized with a latency of approximately one second. While the current implementation provides a functional experience, considering that the human visual system is relatively tolerant of delays up to about 100 milliseconds [18], further improvements could significantly enhance the Semantic See-through experience.

Transformation speed is contingent not only on computational resources but also on the model’s inference speed, which generally exists in a trade-off with accuracy. High-accuracy AI models, including those for image captioning and generation, tend to require more computational resources and processing time, leading to increased inference times as accuracy improves [2]. Therefore, for our system, which demands real-time performance, it is crucial to strike an appropriate balance between accuracy and processing time. The parameters that determine this optimal design ultimately depend on human perceptual characteristics: specifically, the maximum latency that still allows for a suitably enhanced experience.

In the design process, it must be noted that efforts to reduce latency can themselves impact the preservation of semantic meaning. For example, in the prototype presented in this paper, we made several adjustments to tune for accuracy and inference time: the models were selected for their lightweight nature, text length was limited to 20–50 words, and the number of inference steps for image generation was set to four. While these measures successfully reduced the latency to approximately one second, they may have introduced significant information loss. This design difficulty places a significant constraint on the range of models that can be applied.

From the perspective of ensuring real-time performance, the current implementation does not incorporate the preservation of temporal meaning. In other words, each image is processed independently, and the temporal relationships between images are lost. This is because when an entire image is compressed into a single static feature in a standard model, the temporal correlation and motion information between successive frames in a video are lost [22]. As a result, in prototypes, what should be described as “A person is putting down an apple,” for example, might be incorrectly captioned as “A person is picking up an apple” because the movement of the object is not captured. However, it may be possible to better capture the meaning of the motion by externally adding temporal information, such as converting “an apple” into “an apple which a girl tossed to me.” In the future, outputs that retain temporal information could be achieved by using an image-captioning model that takes video as input [23] or a generative model that generates video from text [32].

7.3.2 Framework limitations. Semantic See-through Goggles is designed to make, as a first-person experience, the Linguistic VR that is latent in the mediation of sensory information by language, and in practice it is configured as a video see-through goggle that serially connects Image-to-Text (I2T) and Text-to-Image (T2I). In what follows, we explain points that require care when linking this

configuration to Linguistic VR. For clarity, we use “reconstruction” to denote the step in which under-specified linguistic content is concretized into a fully visual depiction by the system.

First, in the linguistic mediation of a scene, the ultimately imagined sensory content does not arise as a vivid image comparable to a sensory stimulus; rather, it remains partially linguistic and is characteristically indeterminate. In our framework, however, linguistic information is fully visualized, and therefore the missing details must be reconstructed by the AI. Yet when a scene is recalled from language in ordinary imagination, the insufficiency of information remains tacit and is not necessarily made concrete on the spot. In this sense, there is a gap between the final image formed in cases such as “conveying a scene by words and imagining it in one’s head” and the output of Semantic See-through Goggles. We call this the “reconstruction gap,” namely, the difference between language-based mental imagery that tolerates under-specification and the system’s obligation to commit to pixel-level specifics.

Nevertheless, it is indeed the case that information not directly verbalized is supplemented by common sense and bias, allowing imagination to take shape; the AI’s redrawing in this project can be sufficiently understood as an extension of that process. Moreover, when a human, on the basis of information obtained through linguistic mediation, actually undertakes the act of drawing and creates an image, something equivalent to the system’s re-creation of scene details likewise occurs. Thus, the AI’s reconstruction operationalizes a completion step that humans routinely perform implicitly.

Although the present work focuses on vision, the applicability of linguistic mediation to other modalities and to multimodal sensory information requires more detailed examination. For example, the ease with which content can be converted into linguistic information differs by modality; in this sense, modality itself is among the conditions that determine what is and is not carried when words are used as the medium. It is, for instance, well known that touch has relatively sparse descriptive vocabulary. Accordingly, any multimodal extension must analyze modality-specific “translatability into language” and specify how losses in I2T are to be compensated—whether by T2I, text-to-audio, or text-to-haptics pipelines.

8 Conclusion

In this paper, we proposed an eyeglass-based framework, “Semantic See-through Goggles,” that serially inserts Image-to-Text (I2T) and Text-to-Image (T2I) transformations between the head-camera input of a video see-through HMD and its display output. We do so by superimposing a working assumption latent in linguistic mediation—that “realities that are compressed into the same sentence are virtual”—onto a working assumption that grounds much of VR—that “realities that yield the same sensory stimulation are virtual.” In this way, the device foregrounds, from a first-person perspective, an aspect of language as virtual reality.

Experiencing the system—where only what “passes as language” is allowed through vision, and action proceeds in physical space—offers a method for subjectively apprehending AI-mediated visual information, especially the processes of selection, summarization, averaging, and reconstruction. It also provides an opportunity to

understand, in connection with one's bodily sense, the issues that arise inevitably when sensory information is mediated by language.

We believe this study connects a foundational question in VR research—"What is the virtuality of reality?"—with a problem of linguistic communication and rupture—"What can language convey?" In doing so, it contributes to critical understanding and practical resilience in contemporary circumstances in which AI and humans co-organize recognition under meaning.

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References

- [1] Peter Anderson, Xiaodong He, Chris Buehler, Damien Teney, Mark Johnson, Stephen Gould, and Lei Zhang. 2018. Bottom-up and top-down attention for image captioning and visual question answering. In *Proceedings of the IEEE conference on computer vision and pattern recognition*. 6077–6086.
- [2] Phyllis Ang, Bhuwan Dhingra, and Lisa Wu Wills. 2022. Characterizing the efficiency vs. accuracy trade-off for long-context NLP models. *arXiv preprint arXiv:2204.07288* (2022).
- [3] Ronald T Azuma. 1997. A survey of augmented reality. *Presence: teleoperators & virtual environments* 6, 4 (1997), 355–385.
- [4] Michael Bajura, Henry Fuchs, and Ryutarou Ohbuchi. 1992. Merging virtual objects with the real world: Seeing ultrasound imagery within the patient. *ACM SIGGRAPH Computer Graphics* 26, 2 (1992), 203–210.
- [5] Solon Barocas and Andrew D Selbst. 2016. Big data's disparate impact. *Calif. L. Rev.* 104 (2016), 671.
- [6] Lawrence W Barsalou. 1999. Perceptual symbol systems. *Behavioral and brain sciences* 22, 4 (1999), 577–660.
- [7] Raffaella Bernardi, Ruket Cakici, Desmond Elliott, Aykut Erdem, Erkut Erdem, Nazli Ikizler-Cinbis, Frank Keller, Adrian Muscat, and Barbara Plank. 2016. Automatic description generation from images: A survey of models, datasets, and evaluation measures. *Journal of Artificial Intelligence Research* 55 (2016), 409–442.
- [8] Shruti Bhargava and David Forsyth. 2019. Exposing and correcting the gender bias in image captioning datasets and models. *arXiv preprint arXiv:1912.00578* (2019).
- [9] Abeba Birhane and Fred Cummins. 2019. Algorithmic injustices: Towards a relational ethics. *arXiv preprint arXiv:1912.07376* (2019).
- [10] Tolga Bolukbasi, Kai-Wei Chang, James Y Zou, Venkatesh Saligrama, and Adam T Kalai. 2016. Man is to computer programmer as woman is to homemaker? debiasing word embeddings. *Advances in neural information processing systems* 29 (2016).
- [11] Pierre Bourdieu. 1991. Language and symbolic power. *Polity* (1991).
- [12] Joy Buolamwini and Timnit Gebru. 2018. Gender shades: Intersectional accuracy disparities in commercial gender classification. In *Conference on fairness, accountability and transparency*. PMLR, 77–91.
- [13] Aylin Caliskan, Joanna J Bryson, and Arvind Narayanan. 2017. Semantics derived automatically from language corpora contain human-like biases. *Science* 356, 6334 (2017), 183–186.
- [14] Daniel Cer, Yinfei Yang, Sheng-yi Kong, Nan Hua, Nicole Limtiaco, Rhomni St John, Noah Constant, Mario Guajardo-Cespedes, Steve Yuan, Chris Tar, et al. 2018. Universal sentence encoder for English. In *Proceedings of the 2018 conference on empirical methods in natural language processing: system demonstrations*. 169–174.
- [15] Jacob Cohen. 2013. *Statistical power analysis for the behavioral sciences*. routledge.
- [16] Emily A Cooper. 2023. The perceptual science of augmented reality. *Annual Review of Vision Science* 9, 1 (2023), 455–478.
- [17] Kate Crawford. 2021. *The atlas of AI: Power, politics, and the planetary costs of artificial intelligence*. Yale University Press.
- [18] James R. Dabrowski and Ethan V. Munson. 2001. Is 100 Milliseconds Too Fast?. In *CHI '01 Extended Abstracts on Human Factors in Computing Systems* (Seattle, Washington) (CHI EA '01). Association for Computing Machinery, New York, NY, USA, 317–318. doi:10.1145/634067.634255
- [19] Finale Doshi-Velez and Been Kim. 2017. Towards a rigorous science of interpretable machine learning. *arXiv preprint arXiv:1702.08608* (2017).
- [20] Julia Dressel and Hany Farid. 2018. The accuracy, fairness, and limits of predicting recidivism. *Science advances* 4, 1 (2018), eaao5580.
- [21] Luciano Floridi and Massimo Chiriatti. 2020. GPT-3: Its nature, scope, limits, and consequences. *Minds and machines* 30, 4 (2020), 681–694.
- [22] Lianli Gao, Zhao Guo, Hanwang Zhang, Xing Xu, and Heng Tao Shen. 2017. Video Captioning With Attention-Based LSTM and Semantic Consistency. *IEEE Transactions on Multimedia* 19, 9 (2017), 2045–2055. doi:10.1109/TMM.2017.2729019
- [23] Lianli Gao, Zhao Guo, Hanwang Zhang, Xing Xu, and Heng Tao Shen. 2017. Video Captioning With Attention-Based LSTM and Semantic Consistency. *IEEE Transactions on Multimedia* 19 (2017), 2045–2055. doi:10.1109/TMM.2017.2729019
- [24] Kasturi Ghosh and D Ghosh. 2025. Who am AI? Deconstructing Identity in the Age of Advanced Artificial Intelligence.
- [25] Hila Gonen and Yoav Goldberg. 2019. Lipstick on a pig: Debiasing methods cover up systematic gender biases in word embeddings but do not remove them. *arXiv preprint arXiv:1903.03862* (2019).
- [26] Ian J Goodfellow, Jean Pouget-Abadie, Mehdi Mirza, Bing Xu, David Warde-Farley, Sherjil Ozair, Aaron Courville, and Yoshua Bengio. 2014. Generative adversarial nets. *Advances in neural information processing systems* 27 (2014).
- [27] Jan Gugenheimer, Evgeny Stemasov, Julian Frommel, and Enrico Rukzio. 2017. Sharevr: Enabling co-located experiences for virtual reality between hmd and non-hmd users. In *Proceedings of the 2017 CHI conference on human factors in computing systems*. 4021–4033.
- [28] Jan Gugenheimer, Evgeny Stemasov, Harpreet Sareen, and Enrico Rukzio. 2018. A Demonstration of FaceDisplay: Asymmetric Multi-User Interaction for Mobile VR. In *2018 IEEE Conference on Virtual Reality and 3D User Interfaces (VR)*. IEEE, 753–754.
- [29] Melissa Hall, Laurens van der Maaten, Laura Gustafson, Maxwell Jones, and Aaron Adcock. 2022. A systematic study of bias amplification. *arXiv preprint arXiv:2201.11706* (2022).
- [30] Stevan Harnad. 1990. The symbol grounding problem. *Physica D: Nonlinear Phenomena* 42, 1-3 (1990), 335–346.
- [31] Zellig S Harris. 1954. Distributional structure. *Word* 10, 2-3 (1954), 146–162.
- [32] Jonathan Ho, William Chan, Chitwan Saharia, Jay Whang, Ruiqi Gao, A. Gritsenko, Diederik P. Kingma, Ben Poole, Mohammad Norouzi, David J. Fleet, and Tim Salimans. 2022. Imagen Video: High Definition Video Generation with Diffusion Models. *ArXiv abs/2210.02303* (2022). doi:10.48550/arXiv.2210.02303
- [33] Xiao Hu, Haobo Wang, Anirudh Vegesana, Somesh Dube, Kaiwen Yu, Gore Kao, Shuo-Han Chen, Yung-Hsiang Lu, G. Thiruvathukal, and Ming Yin. 2020. Crowdsourcing Detection of Sampling Biases in Image Datasets. *Proceedings of The Web Conference 2020* (2020). doi:10.1145/3366423.3380063
- [34] Shaohan Huang, Li Dong, Wenhui Wang, Yaru Hao, Saksham Singhal, Shuming Ma, Tengchao Lv, Lei Cui, Owais Khan Mohammed, Barun Patra, et al. 2023. Language is not all you need: Aligning perception with language models. *Advances in Neural Information Processing Systems* 36 (2023), 72096–72109.
- [35] Jungo Kasai, Keisuke Sakaguchi, Lavinia Dunagan, Jacob Daniel Morrison, Roman Le Bras, Yejin Choi, and Noah A. Smith. 2021. Transparent Human Evaluation for Image Captioning. *ArXiv abs/2111.08940* (2021). doi:10.18653/v1/2022.naacl-main.254
- [36] Diederik P Kingma and Max Welling. 2013. Auto-encoding variational bayes. *arXiv preprint arXiv:1312.6114* (2013).
- [37] Hubert Klocker, Carl Aigner, and Peter Weibel. 1997. *Alfons Schilling: Ich/Auge/Welt-The Art of Vision*. Springer.
- [38] Matt Kusner, Yu Sun, Nicholas Kolkin, and Kilian Weinberger. 2015. From word embeddings to document distances. In *International conference on machine learning*. PMLR, 957–966.
- [39] S Lee, John H. Xin, and Stephen Westland. 2005. Evaluation of Image Similarity by Histogram Intersection. *Color Research and Application* 30 (2005), 265–274. https://api.semanticscholar.org/CorpusID:54697123
- [40] André Lefevre. 2016. *Translation, rewriting, and the manipulation of literary fame*. Routledge.
- [41] Junnan Li, Dongxu Li, Caiming Xiong, and Steven Hoi. 2022. Blip: Bootstrapping language-image pre-training for unified vision-language understanding and generation. In *International conference on machine learning*. PMLR, 12888–12900.
- [42] Jun Li, Chenyang Zhang, Wei Zhu, and Yawei Ren. 2024. A Comprehensive Survey of Image Generation Models Based on Deep Learning. *Annals of Data Science* (2024), 1–30.
- [43] Zhiheng Li and Chenliang Xu. 2021. Discover the Unknown Biased Attribute of an Image Classifier. *2021 IEEE/CVF International Conference on Computer Vision (ICCV)* (2021), 14950–14959. doi:10.1109/ICCV48922.2021.01470
- [44] Simian Luo, Yiqin Tan, Longbo Huang, Jian Li, and Hang Zhao. 2023. Latent Consistency Models: Synthesizing High-Resolution Images with Few-Step Inference. *arXiv:2310.04378 [cs.CV]*
- [45] Burak Makav and V. Kılıç. 2019. Smartphone-based Image Captioning for Visually and Hearing Impaired. *2019 11th International Conference on Electrical and Electronics Engineering (ELECO)* (2019), 950–953. doi:10.23919/ELECO47770.2019.8990395
- [46] Steve Mann. 2004. Continuous lifelong capture of personal experience with EyeTap. In *Proceedings of the the 1st ACM workshop on Continuous archival and retrieval of personal experiences*. 1–21.
- [47] Steve Mann, Tom Furness, Yu Yuan, Jay Iorio, and Zixin Wang. 2018. All reality: Virtual, augmented, mixed (x), mediated (x, y), and multim mediated reality. *arXiv*

- preprint arXiv:1804.08386 (2018).
- [48] Ludovica Marinucci, Claudia Mazzuca, and Aldo Gangemi. 2023. Exposing implicit biases and stereotypes in human and artificial intelligence: state of the art and challenges with a focus on gender. *AI & SOCIETY* 38, 2 (2023), 747–761.
 - [49] David Marr. 2010. *Vision: A computational investigation into the human representation and processing of visual information*. MIT press.
 - [50] Paul Milgram and Fumio Kishino. 1994. A taxonomy of mixed reality visual displays. *IEEE TRANSACTIONS on Information and Systems* 77, 12 (1994), 1321–1329.
 - [51] Edwin G Ng, Bo Pang, Piyush Sharma, and Radu Soricut. 2020. Understanding guided image captioning performance across domains. *arXiv preprint arXiv:2012.02339* (2020).
 - [52] A Michael Noll. 1994. The beginnings of computer art in the United States: A memoir. *Leonardo* (1994), 39–44.
 - [53] Eirini Ntoutsi, Pavlos Fafalos, Ujwal Gadiraju, Vasileios Iosifidis, Wolfgang Nejdl, Maria-Esther Vidal, Salvatore Ruggieri, Franco Turini, Symeon Papadopoulos, Emmanouil Krasnakis, et al. 2020. Bias in data-driven artificial intelligence systems—An introductory survey. *Wiley Interdisciplinary Reviews: Data Mining and Knowledge Discovery* 10, 3 (2020), e1356.
 - [54] Mayu Otani, Riku Togashi, Yu Sawai, Ryosuke Ishigami, Yuta Nakashima, Esa Rahtu, Janne Heikkilä, and Shin'ichi Satoh. 2023. Toward verifiable and reproducible human evaluation for text-to-image generation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 14277–14286.
 - [55] Ville Paananen, Jonas Oppenlaender, and Aku Visuri. 2023. Using text-to-image generation for architectural design ideation. *International Journal of Architectural Computing* (2023), 14780771231222783.
 - [56] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, et al. 2021. Learning transferable visual models from natural language supervision. In *International conference on machine learning*. PmlR, 8748–8763.
 - [57] Aditya Ramesh, Prafulla Dhariwal, Alex Nichol, Casey Chu, and Mark Chen. 2022. Hierarchical text-conditional image generation with clip latents. *arXiv preprint arXiv:2204.06125* 1, 2 (2022), 3.
 - [58] Aditya Ramesh, Mikhail Pavlov, Gabriel Goh, Scott Gray, Chelsea Voss, Alec Radford, Mark Chen, and Ilya Sutskever. 2021. Zero-shot text-to-image generation. In *International conference on machine learning*. Pmlr, 8821–8831.
 - [59] N Reimers. 2019. Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks. *arXiv preprint arXiv:1908.10084* (2019).
 - [60] Katja Rogers, Sukran Karaosmanoglu, Dennis Wolf, Frank Steinicke, and Lennart E Nacke. 2021. A best-fit framework and systematic review of asymmetric gameplay in multiplayer virtual reality games. *Frontiers in Virtual Reality* 2 (2021), 694660.
 - [61] Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Björn Ommer. 2022. High-resolution image synthesis with latent diffusion models. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 10684–10695.
 - [62] Olga Russakovsky, Jia Deng, Hao Su, Jonathan Krause, Sanjeev Satheesh, Sean Ma, Zhiheng Huang, Andrej Karpathy, Aditya Khosla, Michael Bernstein, et al. 2015. Imagenet large scale visual recognition challenge. *International journal of computer vision* 115 (2015), 211–252.
 - [63] Pierre Sachse, Ursula Beermann, Markus Martini, Thomas Maran, Markus Domeier, and Marco R Furtner. 2017. “The world is upside down”—The Innsbruck goggle experiments of Theodor Erisman (1883–1961) and Ivo Kohler (1915–1985). *Cortex* 92 (2017), 222–232.
 - [64] Chitwan Saharia, William Chan, Saurabh Saxena, Lala Li, Jay Whang, Emily L Denton, Kamyar Ghasemipour, Raphael Gontijo Lopes, Burcu Karagol Ayan, Tim Salimans, et al. 2022. Photorealistic text-to-image diffusion models with deep language understanding. *Advances in neural information processing systems* 35 (2022), 36479–36494.
 - [65] Himanshu Sharma and Devanand Padha. 2023. A comprehensive survey on image captioning: from handcrafted to deep learning-based techniques, a taxonomy and open research issues. *Artificial Intelligence Review* 56, 11 (2023), 13619–13661.
 - [66] Ryan Steed and Aylin Caliskan. 2021. Image representations learned with unsupervised pre-training contain human-like biases. In *Proceedings of the 2021 ACM conference on fairness, accountability, and transparency*. 701–713.
 - [67] George M Stratton. 1897. Vision without inversion of the retinal image. *Psychological review* 4, 4 (1897), 341.
 - [68] Ivan E Sutherland. 1968. A head-mounted three dimensional display. In *Proceedings of the December 9-11, 1968, fall joint computer conference, part I*. 757–764.
 - [69] Ivan E Sutherland et al. 1965. The ultimate display. In *Proceedings of the IFIP Congress*, Vol. 2. New York, 506–508.
 - [70] Yunlong Tang, Junjia Guo, Pinxin Liu, Zhiyuan Wang, Hang Hua, Jia-Xing Zhong, Yunzhong Xiao, Chao Huang, Luchuan Song, Susan Liang, et al. 2025. Generative ai for cel-animation: A survey. *arXiv preprint arXiv:2501.06250* (2025).
 - [71] Antonio Torralba and Alexei A Efros. 2011. Unbiased look at dataset bias. In *CVPR 2011. IEEE*, 1521–1528.
 - [72] Alasdair Tran, A. Mathews, and Lexing Xie. 2020. Transform and Tell: Entity-Aware News Image Captioning. *2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)* (2020), 13032–13042. doi:10.1109/CVPR42600.2020.01305
 - [73] Jesse Vig. 2019. Visualizing Attention in Transformer-Based Language Representation Models. *ArXiv abs/1904.02679* (2019).
 - [74] Oriol Vinyals, Alexander Toshev, Samy Bengio, and Dumitru Erhan. 2015. Show and tell: A neural image caption generator. In *Proceedings of the IEEE conference on computer vision and pattern recognition*. 3156–3164.
 - [75] Sanchayan Vivekananthan. 2024. Comparative analysis of generative models: Enhancing image synthesis with vaes, gans, and stable diffusion. *arXiv preprint arXiv:2408.08751* (2024).
 - [76] Chiu-Hsuan Wang, Seraphina Yong, Hsin-Yu Chen, Yuan-Syun Ye, and Liwei Chan. 2020. Hmd light: Sharing in-vr experience via head-mounted projector for asymmetric interaction. In *Proceedings of the 33rd Annual ACM Symposium on User Interface Software and Technology*. 472–486.
 - [77] Xingjiao Wu, Luwei Xiao, Yixuan Sun, Junhang Zhang, Tianlong Ma, and Liang He. 2022. A survey of human-in-the-loop for machine learning. *Future Generation Computer Systems* 135 (2022), 364–381.
 - [78] Kelvin Xu, Jimmy Ba, Ryan Kiros, Kyunghyun Cho, Aaron Courville, Ruslan Salakhudinov, Rich Zemel, and Yoshua Bengio. 2015. Show, attend and tell: Neural image caption generation with visual attention. In *International conference on machine learning*. PMLR, 2048–2057.
 - [79] Richard Zhang, Phillip Isola, Alexei A Efros, Eli Shechtman, and Oliver Wang. 2018. The unreasonable effectiveness of deep features as a perceptual metric. In *Proceedings of the IEEE conference on computer vision and pattern recognition*. 586–595.
 - [80] Rolf A Zwaan and Gabriel A Radvansky. 1998. Situation models in language comprehension and memory. *Psychological bulletin* 123, 2 (1998), 162.